Worksheet 2: Transpiration and water transport

Name: _____ Class: _____ No.: _____

Read the topic notes, then answer using the context and simulated results.

Topic notes

01 Transpiration: water loss as vapour

Water **evaporates** from **water films** on mesophyll cell surfaces into **air spaces**. Water vapour **diffuses** down a concentration gradient through **stomata** into the air: **transpiration**.

Mesophyll water films Liquid water	→	Evaporation → air spaces Water vapour forms
→		
Diffusion via stomata Vapour lost to the air		

02 Transpiration pull: transport through xylem

Water loss lowers mesophyll **water potential**. **Osmosis** draws replacement water from neighbouring cells. Water is removed from **xylem vessels**, creating **transpiration pull** and drawing water upwards. With adequate water and other conditions equal, stronger light usually favours **stomatal opening**. A celery stalk is a **petiole** containing **xylem**.

Water potential falls Osmosis replaces water	→	Water leaves xylem Transpiration pull forms
→		
Water moves upwards through xylem vessels		

Part 1 | Context and fair comparison

The gardening club puts celery with leaves in red water. Will light and leaves affect water transport?



1. A comparison setup

How would you plan a comparison setup to find out whether leaves help with water transport?

(Hint: the control group provides a basis for comparison. It helps check whether the change in the experiment is caused by the independent variable rather than other external factors. The control group therefore does not receive the treatment being tested.)

(Hint: how is the control setup different from the experiment setup?)

Reference answer: Compare setup D without leaves with setup B with leaves. Both have a lamp distance of 20 cm and the same other conditions. The main difference is whether leaves are present.

2. Identify the variables

Tick the appropriate factors in each row. You may choose more than one per type.

Variable type	Factors
Independent variable the factor changed on purpose	☑ **Light intensity (光強度)** ☐ Height / average upward speed of the red water trace ☐ Temperature ☐ Humidity (濕度) and air movement ☐ Red water concentration and depth of immersion (浸入深度) ☐ Total leaf area of the setups with leaves ☐ Length and thickness of the celery stalks
Dependent variable the result measured	☐ Light intensity (光強度) ☑ **Height / average upward speed of the red water trace** ☐ Temperature ☐ Humidity (濕度) and air movement ☐ Red water concentration and depth of immersion (浸入深度) ☐ Total leaf area of the setups with leaves ☐ Length and thickness of the celery stalks
Controlled variables the factors kept the same	☐ Light intensity (光強度) ☐ Height / average upward speed of the red water trace ☑ **Temperature** ☑ **Humidity (濕度) and air movement** ☑ **Red water concentration and depth of immersion (浸入深度)** ☑ **Total leaf area of the setups with leaves** ☑ **Length and thickness of the celery stalks**

3. Select the assumptions for a fair comparison (select all that apply)

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

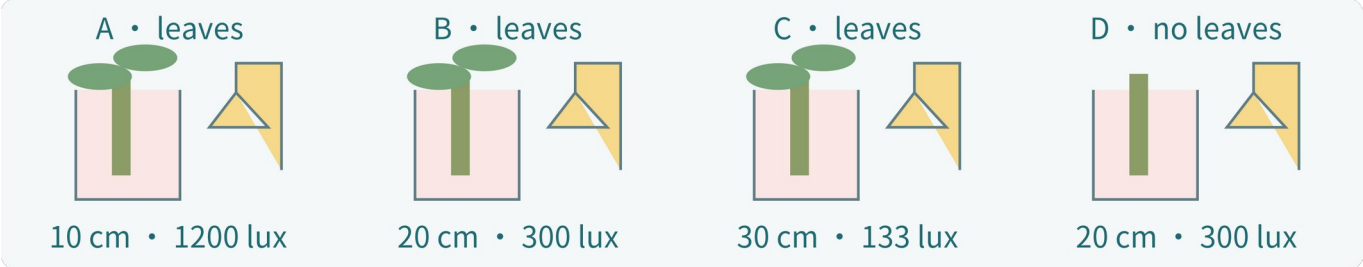
☑ **Changing the lamp distance does not make the setups have different temperatures.**

☑ **Setups A, B and C have the same total leaf area.**

☐ Leaf area does not affect the results, so it does not need to be kept the same.

☑ **Humidity (濕度) and air movement are the same for all setups.**

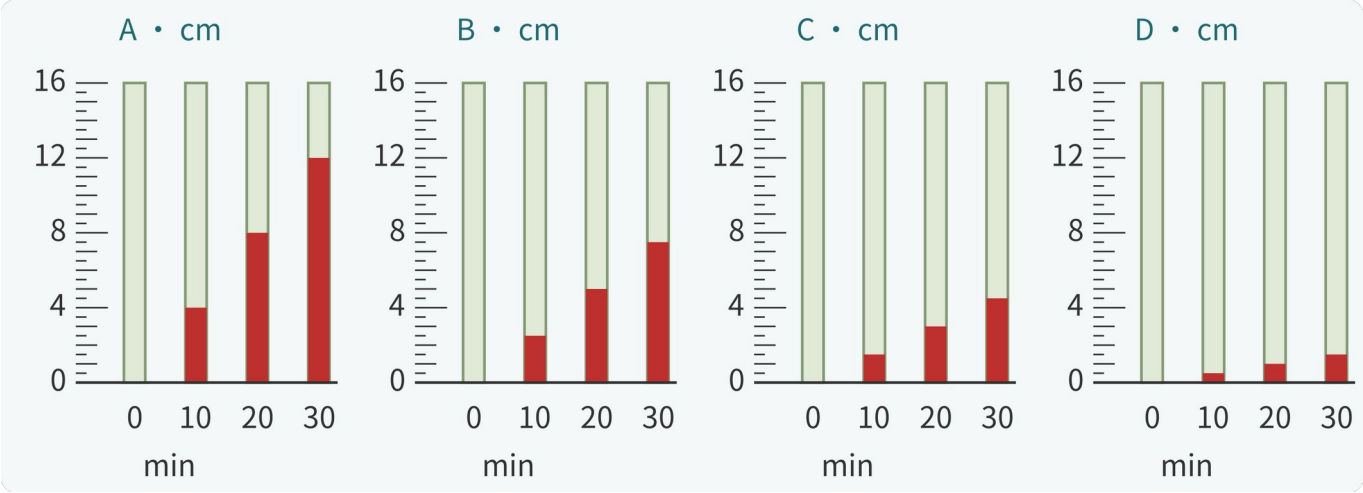
Part 2.1 | Read scales and calculate



All setups: 25°C, equal humidity, air movement and dye concentration, 2 cm immersion, similar petiole dimensions; A/B/C have equal total leaf area.

4. Your measurement record

Read the height of each red trace at each time, to one decimal place (cm).



Time (min)	A	B	C	D
0	**0.0**	**0.0**	**0.0**	**0.0**
10	**4.0**	**2.5**	**1.5**	**0.5**
20	**8.0**	**5.0**	**3.0**	**1.0**
30	**12.0**	**7.5**	**4.5**	**1.5**

5. Calculate average upward speed

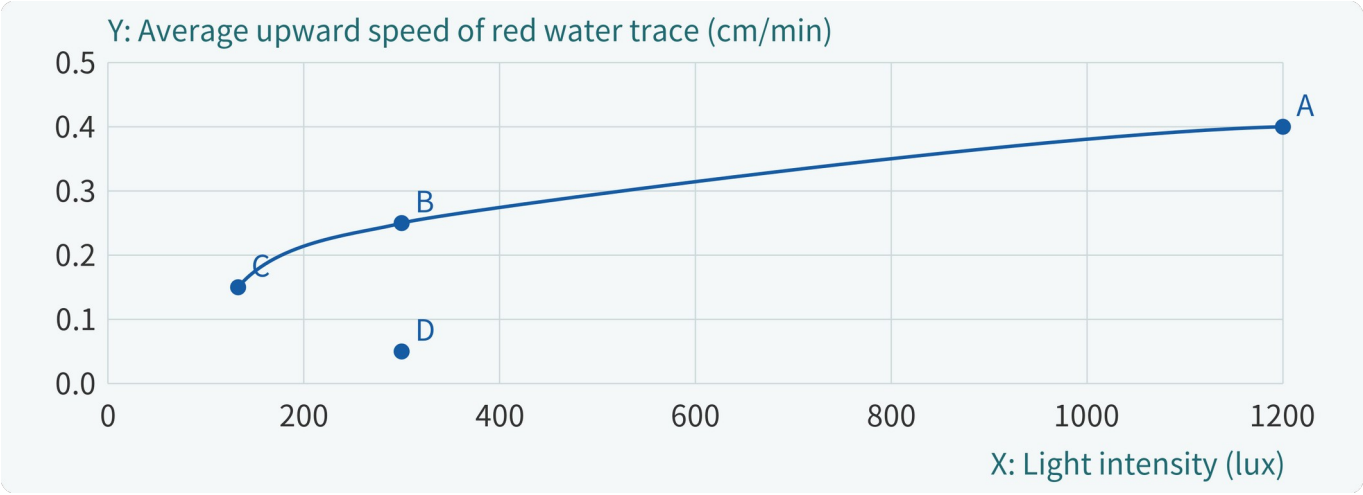
(30 min reading − 0 min reading) ÷ 30; give two decimal places (cm/min).

A	B	C	D
0.40	**0.25**	**0.15**	**0.05**

Part 2.2 | Graph

6. Light intensity and water-trace speed

Plot A/B/C/D using your calculations. Join A/B/C with a smooth trend line; show D separately.



Part 2.3 | Analyse the results

7. How is light intensity (光強度) related to the upward speed of the water trace?

As light intensity (光強度) increases, the average upward speed of the red water trace usually will （**increase** / decrease / show no clear change / show no consistent trend）.

8. What is the role of leaves?

At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is （**faster** / slower / similar）.

9. What is the role of leaves?

This comparison supports the idea that leaves （**help water move upwards** / stop water moving upwards / have no effect on water moving upwards）.

10. What can this experiment tell us?

We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?

☑ **The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率).**

☐ The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率).

☐ If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).

Part 3 | Integrated explanation

11. Using the experimental results and topic notes, explain how light intensity and leaves affect water transport, and explain how transpiration and transpiration pull arise.

Reference answer: With other conditions equal and an adequate water supply, stronger light usually favours stomatal opening. Water evaporates from the water films on mesophyll cell surfaces into the air spaces. Water vapour then diffuses through stomata down a concentration gradient and leaves the leaf, producing transpiration. Water loss lowers the water potential of mesophyll cells, and water from xylem vessels replaces it. Removal of water from the vessels creates transpiration pull, drawing water upwards through xylem vessels. Leaves provide the main surface for transpiration; a setup with leaves usually shows faster upward movement of the water trace than a leafless setup at the same lamp distance. Water-trace speed is an indirect indicator and is not the transpiration rate.